

Empathy or perceived credibility? An empirical study on individual donation behavior in charitable crowdfunding

Study on
individual
donation
behavior

623

Lili Liu

*Department of Management Science and Engineering,
College of Economics and Management,*

Nanjing University of Aeronautics and Astronautics, Nanjing, China, and

Ayong Suh and Christian Wagner

City University of Hong Kong, Kowloon Tong, Hong Kong

Received 21 June 2017
Revised 30 October 2017
3 January 2018
Accepted 3 January 2018

Abstract

Purpose – Researchers have called for the synthesis of divergent perspectives and the development of a theoretical model that examines individuals' donation behavior in charitable crowdfunding. To fill this research gap, the purpose of this paper is to synthesize the literature pertaining to the determinants of donation behavior in charitable crowdfunding. Then, drawing on the stimulus-organism-response framework, the authors develop and test a model that explains individuals' intention to donate to charitable crowdfunding.

Design/methodology/approach – This paper follows a quantitative research approach. An online survey was distributed to collect data from individuals who had experienced charitable crowdfunding. In total, 205 valid responses were received and analyzed.

Findings – First, this study finds that individuals' empathy and the perceived credibility of a charitable crowdfunding project are key determinants for their intention to donate in charitable crowdfunding. Second, the study finds that website quality, transaction convenience, and project content quality influence both empathy and perceived credibility in different ways. Third, it is noteworthy that initiator reputation is positively related to perceived credibility, while project popularity is positively associated with empathy.

Originality/value – This research advances the knowledge of individual donation behavior in charitable crowdfunding. The model can help researchers understand individuals' philanthropic behavior by providing empirical explanations of the interplay between technological and project characteristics, emotional and cognitive states, and individuals' donation behavior. For practitioners, the research suggests appropriate design, launch, and operation strategies to facilitate individuals' donation behavior in charitable crowdfunding.

Keywords Empathy, Perceived credibility, Charitable crowdfunding, Donation behaviour, Project characteristic, Technological characteristic

Paper type Research paper

1. Introduction

Charitable crowdfunding – emerging within the broader landscape of crowdsourcing – describes an open call for monetary donations over a fixed time limit, typically through the medium of web-enabled information systems (Belleflamme *et al.*, 2014; Gerber *et al.*, 2012). In contrast to conventional charity in which a small group of major funders provides money (e.g. bake sales, door knocking, or society events), charitable crowdfunding seeks to obtain external financing from a large number of individuals (the “crowd”) who mainly donate small amounts of money (Lu *et al.*, 2014). Charitable crowdfunding has become a mainstream approach to charity participation, raising 2.85 billion USD globally in 2015 (Massolution, 2015). As a relatively recent

This research was supported in part by Grant No. CityU 11531016 from the Research Grants Council of the Hong Kong SAR awarded to the second author. This research was supported in part by GRF Grant CityU 11507815 from the Research Grants Committee of the Hong Kong Special Administrative Region, awarded to the second and third author.



Internet Research
Vol. 28 No. 3, 2018
pp. 623-651
© Emerald Publishing Limited
1066-2243
DOI 10.1108/IntR-06-2017-0240

and expanding phenomenon, charitable crowdfunding is gaining substantial interest from practitioners and scholars (Choy and Schlagwein, 2016).

Soliciting small amounts of money from a large number of people is not new in charity domains. For example, charitable organizations such as the March of Dimes have been seeking donations from society since 1938 (Whitman, 1972). However, charitable crowdfunding presents a unique context that distinguishes it from traditional charity in many ways. First, while traditional charity intends to carry out large-scale fundraising, charitable crowdfunding focuses on specific and size-limited charity causes (Yang, 2013). Second, in charitable crowdfunding, relatively small contributions are provided by many individuals over a fixed time (normally a few weeks) (Kuppuswamy and Bayus, 2015). Third, charitable crowdfunding provides real-time updates on the communications among participants as well as on the progress of each project, in which potential funders are able to learn the level of support from other funders and the time left before making donation decisions (Kuppuswamy and Bayus, 2015). Fourth, many donations in charitable crowdfunding are made by individuals with few or no social connections to funding seekers (Agrawal *et al.*, 2015). Finally, advanced information technologies (e.g. crowdfunding websites, mobile payment systems) have significantly reduced the coordination and transaction costs in charitable crowdfunding (Choy and Schlagwein, 2016; Moissejev, 2013). In short, charitable crowdfunding has standardized the process of fundraising by implementing information, transaction, and communication tools, thereby challenging existing explanations of donation behavior in conventional acts of charity (Belleflamme *et al.*, 2013).

Despite charitable crowdfunding's popularity and the increasing public attention it is receiving, a great number of charitable crowdfunding campaigns have reportedly failed to achieve their monetary goals within the pre-specified period (Kickstarter, 2013). To understand the success of charitable crowdfunding, research has focused on diverse factors that influence individuals' donation behavior, such as IT affordances (Choy and Schlagwein, 2016), altruism and warm glow motivations (Gleasure and Feller, 2016a), project content (Kuppuswamy and Bayus, 2015), credibility (Liu *et al.*, 2017; Tanaka and Volda, 2016), website quality (Kuo and Wu, 2014), and social networks (Agrawal *et al.*, 2015). Given that researchers have investigated various factors with different focuses, there is a need for a synthesis of divergent perspectives and the development of a theoretical model that systematically examines individual donation behavior in charitable crowdfunding (Feller *et al.*, 2013). However, relatively little attention has been paid to the synthesis of existing knowledge; the question of why people donate for charitable crowdfunding has not been fully answered.

To fill this research gap, we synthesize the literature pertaining to the determinants of donation behavior in charitable crowdfunding. Then, drawing on the stimulus-organism-response (S-O-R) framework, we develop and test a model that explains an individual's intention to donate in charitable crowdfunding. The S-O-R framework serves as a theoretical platform through which we consolidate diverse factors related to the aspects of the project, technology, and individuals' emotional and cognitive states. By linking the subset of underlying psychosocial mechanisms applicable to charitable crowdfunding, this study seeks to expand our knowledge of individuals' donation behavior in charitable crowdfunding.

For academia, this research contributes to advancing our knowledge of individual donation behavior in charitable crowdfunding. Our model can help researchers understand individuals' philanthropic behavior by providing empirical explanations of the interplay between technological and project characteristics, emotional and cognitive states, and individuals' donation behavior. For practitioners, the research suggests appropriate design, launch, and operation strategies to facilitate individuals' donation behavior in charitable crowdfunding.

2. Literature review

2.1 Charitable crowdfunding

Crowdfunding – either charitable or commercial – aims to harness the power of the crowd for many relatively small contributions that together form a budget great enough to execute a project that is unlikely to be realized by traditional means of funding (Lambert and Schwienbacher, 2010). Crowdfunding has matured into a meaningful paradigm for fundraising that is projected to raise \$1 trillion in 2025 (Crowdsourcing, 2012). The previous literature identifies four primary types of crowdfunding, based on what funders receive in exchange for their contribution (Beaulieu *et al.*, 2015; Belleflamme *et al.*, 2014; Gleasure and Feller, 2016b; Groebner *et al.*, 2011; Haas *et al.*, 2014; Mollick, 2014; Salisbury *et al.*, 2001): equity-based crowdfunding (share equities), reward-based crowdfunding (receive a product, service, or other non-monetary rewards), lending-based crowdfunding (share a particular interest rate), and donation-based crowdfunding (or charitable crowdfunding), where the funders receive no material reward. Since each type of crowdfunding features a distinctive operational mode, these four types of crowdfunding are usually analyzed separately (Beier and Wagner, 2015). In this study, we focus on investigating charitable crowdfunding.

The term “charity” commonly describes “the giving of aid to the needy” (Merriam-Webster, 2015). More specifically, donation in charitable crowdfunding refers to the giving of monetary aid to the needy. In general, a charitable crowdfunding project involves three parties: the project initiator, who proposes the idea to be funded; funders whose donations support the project; and platforms, which bring project initiators and funders together to launch the project (Moisseyev, 2013). Charitable crowdfunding platforms, such as Kiva, Chuffed, and Pledge, provide opportunities for any initiator to launch a project and to request a certain amount of money that needs to be raised within a fixed duration (Mollick, 2012). The two largest social network sites (SNSs) in Mainland China (Weibo and WeChat) also serve as charitable crowdfunding platforms that allow users to raise money for charitable purposes. For instance, “weigongyi” in Weibo and “qingsongchou” in WeChat are two popular charity fundraising channels. Launching a charitable crowdfunding project on SNSs is a fast and far-reaching way to broadcast information to the large pool of users and to build widespread support (Lambert and Schwienbacher, 2010), which maximizes the chances of its success (Moisseyev, 2013).

2.2 S-O-R framework

Intention to donate is a crucial factor that signifies charitable crowdfunding success yet has not received in-depth investigation. Our literature review reveals that diverse factors – technological characteristics, crowdfunding project characteristics, individuals’ emotional and cognitive states – are related to individuals’ behavioral intention to donate in charitable crowdfunding (see Table A1). To consolidate factors that are associated with an individual’s intention to donate, this study employs the S-O-R framework (Mummalaeni, 2005). This framework posits that cues perceived in the situated environment (stimuli) trigger one’s internal evaluation (organism), which subsequently brings about positive or negative behaviors in relation to the organism (response) (Jiang *et al.*, 2010). We believe that the S-O-R framework is suitable to consolidate the factors examined in the previous studies because it serves as a theoretical guide to synthesizing divergent perspectives on stimuli, organism, and response in charitable crowdfunding context. On the basis of the S-O-R framework, we develop a model that explores how technological and project characteristics (stimuli) induce individuals’ empathy and their perception of a project’s credibility (organism), which in turn affect their intention to donate (response) in charitable crowdfunding.

2.2.1 Organism: empathy and perceived credibility. Charitable crowdfunding denotes the intersection between conventional charity and online behavior. On the one hand, charitable crowdfunding can be framed as a type of philanthropy (Gerber and Hui, 2013). Hence, some

of the key factors identified in philanthropy are expected to play a role in crowdfunding. In conventional charitable donation projects, individuals who donate money frequently cite altruistic reasons to explain their willingness to donate (Choy and Schlagwein, 2016; Steele *et al.*, 2008). For instance, researchers have found that individuals are stimulated to donate money because of their feelings of empathy for specific crowdfunding projects (Gerber *et al.*, 2012; Rick *et al.*, 2008). In this context, empathy pertains to the extent to which an individual feels compassion for the particular target (i.e. those seeking help), which represents the individuals' emotional state (Hoffman, 2001).

On the other hand, IS research has found that online behavior is facilitated by perceptions of the source's credibility (Cheung *et al.*, 2009). In charitable crowdfunding, individuals' monetary donations are made without expectation of material rewards. Therefore, we expect concerns about credibility to be relevant for the potential funders. Those who fund in charitable crowdfunding are concerned about whether their donation will be abused for another purpose. Thus, before donating money, individuals will evaluate whether a project is credible. Credibility is defined here as a perceptual variable of crowdfunding projects rather than as an objective measure of such projects, which represents individuals' cognitive state (Kim *et al.*, 2016). In other words, credibility is a property that is judged by the individuals who participate in crowdfunding rather than a property of a crowdfunding project *per se* (Freeman and Spyridakis, 2004). Accordingly, we identify empathy (emotional state) and perceived credibility (cognitive state) as predictors of intention to donate in this study.

2.2.2 Stimulus: environmental cues. Environmental cues have been found to influence donation behavior in that individuals may experience the intention to donate money when they are stimulated by certain circumstantial factors while reviewing crowdfunding projects (Bennett, 2009; Moissejev, 2013). Previous research has suggested that technological characteristics and crowdfunding project characteristics are important factors in crowdfunding (Belleflamme *et al.*, 2014; Gerber and Hui, 2013; Gerber *et al.*, 2012; Moissejev, 2013; Mollick, 2012). In this study, technological characteristics are represented by website quality (Everard and Galletta, 2005) and transaction convenience (Belleflamme *et al.*, 2014; Gerber and Hui, 2013). Huber (1990) suggests that when new technology is adopted or when we make a better use of the capability of existing technology, it is important to reassess the effects of the technology. In the charitable crowdfunding context, individuals' donation behavior may be affected by the implementation of new technology that facilitates transactions (e.g. a mobile payment system), which lowers the effort of giving and makes smaller donations worthwhile. In addition, the website quality of a crowdfunding platform (e.g. security) may affect individuals' perceived credibility of a project on that platform, which has long been a challenge for online money spending (Hoffman *et al.*, 1999). Previous research has identified crowdfunding project characteristics as those project attributes that relate to the funding decision; these attributes are initiator reputation (Gerber and Hui, 2013), popularity of crowdfunding project (Lu *et al.*, 2014), and crowdfunding project content quality (Park and Kim, 2003).

The environmental cues have been discussed in previous crowdfunding literature (Belleflamme *et al.*, 2014; Gerber and Hui, 2013; Gerber *et al.*, 2012; Moissejev, 2013; Mollick, 2012). However, there is little knowledge about how these characteristics jointly trigger donation behavior. Collectively, the characteristics stand for the many facets of environmental cues in the context of charitable crowdfunding. Thus, this study extends previous research by investigating whether these cues and their corresponding attributes are major catalysts in increasing the empathy for and perceived credibility of a charitable crowdfunding project, which in turn determine the intention to donate money. In doing so, we seek to investigate the relative importance of the two characteristics and their corresponding attributes with respect to their effect on individuals' donation behavior.

3. Research model and hypotheses

On the basis of the preceding literature review, we propose the research model depicted in Figure 1. Each hypothesis in the proposed model is discussed in detail below.

3.1 *Empathy, perceived credibility, and intention to donate*

Empathy is defined as “an affective state that stems from the apprehension of another’s emotional state or condition” (Eisenberg and Miller, 1987). Empathy relies on automatically activated state matching that produces shared representations and similar emotions (Decety and Jackson, 2006); such state matching is prominent wherever humans attempt to cultivate feelings that are more just and compassionate (Gerdes, 2011). Researchers have found that empathy motivates prosocial behaviors (Batson *et al.*, 1995; Eisenberg and Miller, 1987; Murillo *et al.*, 2016), such as donating money to charitable crowdfunding projects (Lee *et al.*, 2014). Hence, cultivating empathy is a main determinant for intention to donate. Accordingly, we formulate the following hypothesis:

H1. Empathy is positively related to intention to donate.

Perceived credibility, defined in the study as the believability of a crowdfunding project in the judgment of potential funders (O’Keefe, 2015), has been extensively studied in the context of online transactions (Castillo *et al.*, 2011; Gvili and Levy, 2016; Kim *et al.*, 2016). Previous works have indicated that although most of the messages posted on crowdfunding platforms are credible, at times such platforms are also used to spread misinformation and false rumors (Castillo *et al.*, 2011). Hence, while reviewing a crowdfunding project, potential funders often rely on their assessment of the project’s credibility when deciding whether to donate money (Greenberg and Gerber, 2014). We thus propose the following hypothesis:

H2. Perceived credibility is positively related to intention to donate.

3.2 *Technological characteristics*

Previous studies have described website quality as a form of technological characteristic (Wells *et al.*, 2011). Following Wells *et al.*’s (2011) study, website quality is identified as a high-order construct consisting of three sub-dimensions: security, navigability, and visual appeal. When website quality is affirmed, an attitude of trust toward or belief in the information/source on the website can be established (Gregg and Walczak, 2010). Furthermore, a well-designed website interface induces positive emotional and cognitive evaluations of crowdfunding projects. For instance, visual appeal has been found to induce a sharable emotion between audiences (Capota *et al.*, 2007). Thus, a well-designed,

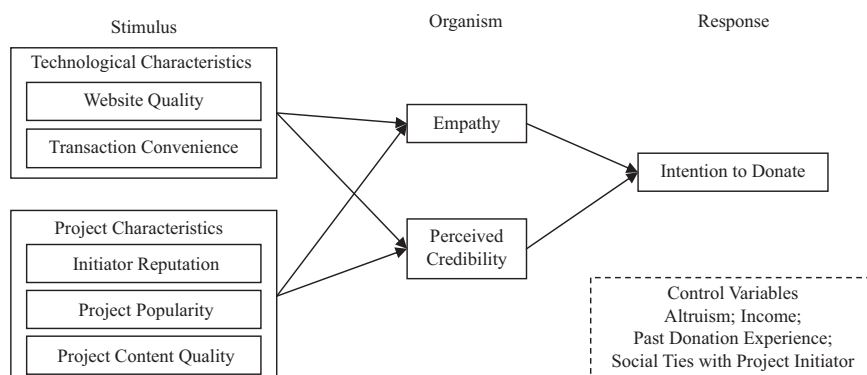


Figure 1.
Research model

high-quality website will increase individuals' empathy for a crowdfunding project, as well as the project's perceived credibility. We therefore propose the following hypotheses:

H3a. Website quality is positively related to empathy.

H3b. Website quality is positively related to perceived credibility.

Transaction convenience is defined as the extent to which it is easy to initiate, authorize, and confirm a transaction (Kim *et al.*, 2010). In crowdfunding, existing online payment systems are adopted to facilitate transactions between initiators and potential funders (Gerber *et al.*, 2012). Transaction convenience has been discussed primarily in the marketing and consumer behavior literature (Jih, 2007), which relates to the elements generating time and place utility for users. For example, in order to transfer money to crowdfunding projects (initiators and potential funders are geographically isolated), users of Weibo are able to pay through their e-banking system, Alipay system, or WeChat payment system, all of which are more convenient than visiting a physical bank. In fact, funding behavior appears instantaneously during or immediately after reading the information of a crowdfunding project (Bennett, 2009). One possible explanation is that a potential funder's attention quickly shifts (Crawford, 2015, p. 11). If there is a long time delay, potential funders are likely to be unwilling to return their attention to the previously viewed crowdfunding project, expend the effort of possibly having to find it again, and learn how to transfer money to it. Furthermore, an individual's intention to donate money is also affected by the circumstances of use, such as time pressure (project deadline) for the payment (Kim *et al.*, 2010). Hence, if a transaction can be accomplished easily, individuals may perceive the project as highly credible and quickly generate empathy toward it, which in turn motivates their funding behavior. We thus propose the following hypotheses:

H4a. Transaction convenience is positively related to perceived credibility.

H4b. Transaction convenience is positively related to empathy.

3.3 Project characteristics

Initiator reputation describes the extent to which a potential funder believes that an initiator is honest and concerned about the funders (Jøsang *et al.*, 2007). This definition corresponds well with the position of researchers (Marsden and Lin, 1982) that a reputation is a quantity of good impressions derived from the underlying website that is globally visible to all members of the network. Prior research has noted that this construct has a positive and direct effect on building positive attitudes (e.g. trust) toward people or objects (Jin *et al.*, 2008). Moreover, a favorable reputation can bring several important benefits to individuals or organizations, for instance, because people rely on reputation information when they choose partners with whom to work, they are more willing to apprehend the feelings of reputable partners and support them (Jøsang *et al.*, 2007). We thus hypothesize the following:

H5a. Initiator reputation is positively related to empathy.

H5b. Initiator reputation is positively related to perceived credibility.

The popularity of a crowdfunding project is represented by the number of retweets, comments, and likes related to it (De Vries *et al.*, 2012). Once launched, crowdfunding projects have the potential to spread to other websites (e.g. Twitter, Weibo, WeChat) and to be viewed by potential funders (Guo *et al.*, 2013). Crowdfunding projects with large numbers of retweets, comments, and likes are popular. Previous research has demonstrated that project popularity serves as a cue and that, as it increases, potential funders' perception of a project's credibility also increases (Jin and Phua, 2014; Westerman *et al.*, 2012). Moreover, others users' involvement in a crowdfunding project (e.g. retweeting the crowdfunding project's

information, liking the project, and commenting on the project) is conducive to generating empathy and positive feelings among future viewers, who may become funders of the project (Bickart and Schindler, 2001). These findings are consistent with previous research on popularity. For example, it is demonstrated that people often buy and download music based on its popularity (Salganik *et al.*, 2006). Therefore, we propose the following hypotheses:

H6a. Project popularity is positively related to empathy.

H6b. Project popularity is positively related to perceived credibility.

In this study, project content quality is defined as the degree to which an individual believes that the information provided about a crowdfunding project is of a high quality (Xu *et al.*, 2013). Because it is often the case that multiple crowdfunding projects are concurrently raising money for similar purposes, individuals seek information about crowdfunding projects that allows them to distinguish those with high credibility from those with low credibility by acquiring more detailed information (e.g. the initiator, aim, anticipated amount of money, available period). Empirical findings have supported the observation that project content quality positively influences the user's perception of a project's credibility (Fung and Lee, 1999). In addition, when a crowdfunding project provide complete, accurate, well-formatted, and timely information, individuals are more likely to generate an emotional resonance (e.g. empathy) with it, based on a deeper understanding of the project (Lee *et al.*, 2008). Thus, we propose:

H7a. Project content quality is positively related to empathy.

H7b. Project content quality is positively related to perceived credibility.

4. Methodology

4.1 Measurement

All the measurement items have been adopted from the literature and modified to reflect the context of charitable crowdfunding. Seven-point Likert scale ranging from 1 (strongly disagree) to 7 (strongly agree) were used to measure the items. Website quality was assessed with items adapted from Gleasure and Feller (2016b), Salisbury *et al.* (2001), Loiacono *et al.* (2007). These items measured three sub-dimensions of the crowdfunding website: security, navigability, and visual appeal. Items for transaction convenience were adapted from Kim *et al.* (2010). Items of initiator reputation were adapted from Jarvenpaa *et al.* (1999). Project popularity items were adapted from Chang *et al.* (2015). Items of project content quality were adapted from the work of Xu *et al.* (2013). Items for perceived credibility were adapted from McCroskey and Teven (1999). Items of Empathy were adapted from Batson *et al.* (1987). Finally, items of intention to donate were adapted from Dodds *et al.* (1991).

Previous research demonstrates that individuals' general tendency to be altruistic differ considerably (Steele *et al.*, 2008). Hence, altruism was treated in this study as a control variable of intention to donate. We measured an individual's inherent altruism using four items adapted from Chen *et al.* (1998). In addition, many donations in charitable crowdfunding were made by individuals who built weak or no social connections with fund seekers (Agrawal *et al.*, 2015). We thus examined individuals' social ties with the project initiator as another control variable of intention to donate. Items for measuring social ties with a project initiator were derived from the work of Chiu *et al.* (2006). Moreover, we also included individuals' income and past donation experience as control variables of intention to donate. A summary of the measurement items and their sources is presented in Table AII.

The questionnaire was originally developed in English and subsequently translated into Chinese by the first author while another bilingual professor conducted the back translation. To ensure the accuracy of translation, three bilingual PhD students helped to

verify that both versions reflected the same content. The instruments were further pretested with a convenience sample of 15 participants, confirming that the wording and instructions were appropriate. According to the pretest, minor revisions were made to the questionnaire.

4.2 Data collection

To test our hypotheses, we collected data from the users of WeChat and Weibo, SNSs in Mainland China. These SNSs allow initiators to directly post charitable crowdfunding projects, recruit funders, and solicit the required amount of money based on an agreed-upon deadline. Although charitable crowdfunding can function without SNSs, it has greatly developed along with the advance of SNSs. It has been argued that use of SNSs may help initiators to reach fundraising targets faster (Moisseyev, 2013). For instance, in 2014, Weibo (Chinese Twitter) announced that within 20 days, 40,000 users collectively donated \$1.5 million for Amyotrophic Lateral Sclerosis patients, an amount almost five times greater than the total amount of donations for this charitable purpose in 2013.

An online survey agency called SO JUMP (www.sojump.com) was commissioned to recruit survey participants, targeting Weibo and WeChat users via an e-mail invitation, from May 12 to May 26, 2016. Respondents were asked to recall the latest charitable crowdfunding project they had read (regardless of whether they had donated money) and to fill out our questionnaire. In two weeks, a total of 403 respondents were approached, of which 230 (57 percent) people completed the questionnaire. Among these respondents, 25 were excluded as they had reported nonexistent charitable crowdfunding projects (we asked each respondent to provide the title of the latest charitable crowdfunding project he or she had read and double-checked the authenticity of each project). Finally, 205 valid responses were collected during this process. Table I lists the demographic information of the respondents. To summarize, 45.85 percent were male and 54.15 percent were female, with ages ranging from 18 to 50; education levels varied from associate degree to doctoral degree, and 56 percent had an income of less than CNY 5,000 per month.

Item	Category	Frequency	Ratio (%)
Gender	Male	94	45.85
	Female	111	54.15
Age	≤20	9	4.39
	21-30	118	57.56
	31-40	55	26.83
	41-50	23	11.22
Education	Associate degree	54	26.35
	Bachelor degree	128	62.44
	Master degree	20	9.76
	Doctoral degree	3	1.45
Income (CNY)	≤2,000	24	11.71
	2,001-5,000	91	44.38
	5,001-8,000	56	27.32
	8,001-15,000	24	11.71
	> 15,000	10	4.88
Past donation experience	Never	48	23.41
	Seldom	64	31.22
	Sometimes	81	39.51
	Frequently	12	5.86

Table I.
Subject demographics

5. Data analysis and results

This study utilized a two-step approach to analyze the research model (Chin *et al.*, 2003). The first step involved the analysis of the measurement model, where reliability, validity, and common method bias were examined. The second step tested the structural model, where structural relationships among latent constructs and hypotheses were assessed, respectively (Chin *et al.*, 2003). SmartPLS 2.0 was used to analyze the data (Ringle *et al.*, 2005).

5.1 Measurement model

5.1.1 Reliability, convergent validity, and discriminant validity. The reliability was verified by using three criteria suggested by Hair *et al.* (2006) and Chin (1998): Cronbach's α should exceed 0.7, construct composite reliabilities (CR) should exceed 0.7, and average variance extracted (AVE) by each construct should exceed 0.5. As shown in Table II, all Cronbach's α s were greater than 0.7 and the CR values of all constructs ranged from 0.89 to 0.96. The AVE values ranged from 0.62 to 0.87 (see Table V). These results suggested that all the constructs had adequate reliability (Fornell and Larcker, 1981). Convergent validity for the reflective constructs was assessed by checking the item loadings on the corresponding constructs, while item weights were examined for the formative construct (website quality). Results in Table II showed that all item loadings of reflective constructs were significant at $p < 0.001$ (all the t -values were greater than 3.29), and almost all item loadings exceeded 0.7, indicating

Construct	Item code	Means	Loadings	t -value	α	CR
Transaction convenience	TC1	5.63	0.89	53.47	0.90	0.93
	TC2	5.50	0.89	45.57		
	TC3	5.26	0.85	34.50		
	TC4	5.58	0.88	37.70		
Initiator reputation	REP1	5.65	0.68	16.40	0.85	0.89
	REP2	4.39	0.71	16.50		
	REP 3	4.40	0.77	16.50		
	REP 4	5.01	0.89	53.78		
	REP 5	5.08	0.88	49.40		
Project popularity	POP1	5.45	0.87	33.98	0.88	0.93
	POP2	5.49	0.93	79.36		
	POP3	5.38	0.90	50.62		
Project content quality	CON1	5.23	0.93	72.65	0.93	0.95
	CON2	5.38	0.94	113.45		
	CON3	5.18	0.93	70.21		
Empathy	EMP1	5.81	0.61	9.23	0.88	0.91
	EMP2	5.89	0.72	13.50		
	EMP3	5.40	0.79	20.36		
	EMP4	5.66	0.86	46.36		
	EMP5	5.40	0.85	36.73		
	EMP6	5.75	0.85	33.20		
Perceived credibility	CRE1	5.32	0.91	54.34	0.94	0.96
	CRE2	5.28	0.91	70.70		
	CRE3	5.27	0.87	45.10		
	CRE4	5.28	0.91	66.08		
	CRE5	5.19	0.88	33.22		
	CRE6	4.99	0.82	27.95		
Intention to donate	INT1	5.16	0.94	89.76	0.92	0.95
	INT2	5.22	0.94	106.07		
	INT3	5.23	0.90	37.01		

Notes: α , Cronbach's alpha; CR, composite reliability

Table II.
Item means and
loadings of reflective
constructs

adequate convergent validity (Fornell and Larcker, 1981). For the formative construct, the weights of the three items were significant (see Table III). Furthermore, the results showed that the loadings for formative items were significant, indicating that all the conditions for convergent validity were ensured (Cenfetelli and Bassellier, 2009). In addition, values for skewness (a measure of the asymmetry of a distribution) and kurtosis (a measure of peakedness of a distribution) were tested using IBM SPSS Statistics 23. The results showed that skewness values ranged from -0.439 to 0.205 while kurtosis values ranged from -0.926 to -0.051, indicating the reasonable normality of the distribution (skewness and kurtosis values should ranged from -1.5 and 1.5) (Brown, 2011; Hair *et al.*, 1998).

Two steps were performed to assess the discriminant validity of the constructs (Fornell and Larcker, 1981). First, the AVE analysis was conducted. If the square root value of the AVE for each construct is greater than the correlations between constructs that are not theoretically related, the discriminant validity is confirmed. As shown in Table V, all the square root values of the AVE (values on the diagonal) exceeded the inter-construct correlations, suggesting good discriminant validity. The study gains further confidence on discriminant validity by examining the cross-loadings, which indicate that items are more strongly related to their own construct than other constructs (see Tables IV and V).

5.1.2 Common method biases. As Podsakoff *et al.* (2003) and Podsakoff and Organ (1986) argued, with self-reported data, common method biases resulting from multiple sources, e.g., consistency motif and social desirability, would potentially show up. In our study, website quality was measured with formative indicators. According to Liang *et al.*'s (2007) statistical approach, in which they also adopted formative measurements, we first performed Harmon one-factor test on the pivotal constructs (Podsakoff and Organ, 1986), including transaction convenience, initiator reputation, project popularity, project content quality, empathy, perceived credibility, and intention to donate. The greatest covariance explained by the one factor was 22.37 percent, indicating that common method bias were unlikely to be a serious issue in this study. Second, following Podsakoff *et al.* (2003) and Liang *et al.* (2007), the variance explained by a common method factor and the substantive factors were compared to determine whether the majority of covariance was explained by the common method factor. In advance of common method bias evaluation, we tested two different versions of the theoretical model. In one version, website quality was considered as a formative construct, while in the other version, it was treated as a reflective construct. As a result, all the significant paths remained and no paths gained or lost statistical significance, advising that regardless of whether website quality was formatively or reflectively measured, there was no qualitative difference (Liang *et al.*, 2007). Therefore, website quality was modeled as a reflective construct when we used SmartPLS 2.0 to examine common method bias.

In light of Liang *et al.*'s (2007) study, each indicator in the research model were transformed to a first-order construct, and then linked to a second-order construct (the first-order construct before conversion). Corresponding indicators were used to measure these first-order constructs (the indicators before conversion). Moreover, following Podsakoff *et al.* (2003) and Williams *et al.*'s (2003) procedure, a common method factor

Table III.
Item means and
loadings of formative
constructs

Construct	Item code	Means	Weights	t-value	Loading	t-value
Website quality	SEC	4.40	0.42	4.54	0.72	13.49
	NAV	5.76	0.56	5.38	0.84	23.09
	VIS	5.24	0.30	2.33	0.78	32.32

Notes: SEC, security; NAV, navigability; VIS, visual appeal

	WQ	TC	REP	POP	CON	EMP	CRE	INT
SEC	0.72	0.39	0.30	0.27	0.33	0.35	0.46	0.48
NAV	0.84	0.52	0.24	0.38	0.36	0.56	0.37	0.36
VIS	0.78	0.38	0.37	0.45	0.44	0.46	0.41	0.41
TC1	0.53	0.89	0.20	0.32	0.34	0.50	0.40	0.51
TC2	0.48	0.89	0.21	0.30	0.34	0.44	0.40	0.53
TC3	0.41	0.85	0.26	0.32	0.34	0.39	0.38	0.57
TC4	0.56	0.88	0.22	0.41	0.36	0.52	0.41	0.48
REP1	0.18	0.23	0.68	0.44	0.54	0.33	0.48	0.34
REP2	0.28	0.19	0.71	0.30	0.37	0.36	0.41	0.34
REP 3	0.23	0.05	0.77	0.29	0.31	0.24	0.21	0.21
REP 4	0.35	0.17	0.89	0.38	0.47	0.38	0.44	0.37
REP 5	0.40	0.27	0.88	0.48	0.54	0.37	0.44	0.45
POP1	0.37	0.28	0.49	0.87	0.52	0.47	0.42	0.50
POP2	0.40	0.34	0.43	0.93	0.52	0.45	0.46	0.50
POP3	0.46	0.42	0.40	0.90	0.54	0.43	0.47	0.54
CON1	0.41	0.39	0.52	0.54	0.93	0.52	0.62	0.58
CON2	0.44	0.35	0.56	0.51	0.94	0.55	0.64	0.58
CON3	0.45	0.36	0.57	0.60	0.93	0.48	0.61	0.56
EMP1	0.33	0.32	0.27	0.32	0.33	0.61	0.32	0.29
EMP2	0.43	0.37	0.36	0.42	0.48	0.72	0.45	0.40
EMP3	0.46	0.38	0.33	0.35	0.38	0.79	0.46	0.43
EMP4	0.52	0.42	0.32	0.39	0.46	0.86	0.45	0.51
EMP5	0.47	0.46	0.35	0.39	0.43	0.85	0.37	0.53
EMP6	0.56	0.53	0.42	0.49	0.51	0.85	0.47	0.64
CRE1	0.47	0.42	0.45	0.49	0.63	0.49	0.91	0.55
CRE2	0.51	0.46	0.46	0.49	0.60	0.49	0.91	0.55
CRE3	0.42	0.37	0.43	0.44	0.60	0.51	0.87	0.50
CRE4	0.44	0.40	0.46	0.43	0.60	0.48	0.91	0.50
CRE5	0.48	0.44	0.53	0.45	0.61	0.47	0.88	0.61
CRE6	0.44	0.29	0.47	0.35	0.52	0.39	0.82	0.50
INT1	0.54	0.52	0.47	0.57	0.62	0.61	0.61	0.94
INT2	0.45	0.56	0.44	0.55	0.57	0.58	0.58	0.94
INT3	0.46	0.57	0.32	0.45	0.50	0.49	0.50	0.90

Notes: WQ, website quality; SEC, security; NAV, navigability; VIS, visual appeal; TC, transaction convenience; REP, initiator reputation; POP, project popularity; CON, project content quality; EMP, empathy; CRE, perceived credibility; INT, intention to donate

Table IV.
Cross-loadings

	AVE	WQ	TC	REP	POP	CON	EMP	CRE	INT
WQ	na	na							
TC	0.77	0.57	0.88						
REP	0.63	0.37	0.25	0.79					
POP	0.81	0.46	0.39	0.49	0.90				
CON	0.87	0.47	0.39	0.59	0.59	0.93			
EMP	0.62	0.60	0.53	0.44	0.50	0.55	0.79		
CRE	0.78	0.52	0.45	0.53	0.50	0.67	0.54	0.89	
INT	0.86	0.52	0.59	0.45	0.57	0.61	0.61	0.61	0.93

Notes: WQ, website quality; TC, transaction convenience; REP, initiator reputation; POP, project popularity; CON, project content quality; EMP, empathy; CRE, perceived credibility; INT, intention to donate; AVE, average variance extracted. The square root of average variance extracted (AVE) is shown on the diagonal of the correlation matrix

Table V.
Discriminant validity

with items of all constructs was created and included into the converted research model. Thus, the variance of each item was explained substantively by its principle construct, as well as by the method factor. Results in Table VI indicated that the substantive factor loading ($R1$) of each item was significant at $p < 0.001$ and was much greater than corresponding method factor loadings ($R2$). All of the method factor loadings were not significant, justifying that common method bias was not a serious concern for this study (Williams *et al.*, 2003). Besides, the common method factor only explained 0.4 percent of the overall variance ($R2^2$), again confirming that common method bias was not a threat to the results.

5.1.3 Multicollinearity. Although the correlations among different constructs were generally moderate or low, several correlations were above 0.6 (e.g. correlation between perceived credibility and intention to donate). To avoid the threat of multicollinearity, the variance inflation factor (VIF) and tolerance values were examined using the regression module of IBM SPSS Statistics 23. The results showed that the VIF values for all the independent variables ranged from 1.217 to 2.291, whereas the tolerance values ranged from

Construct	Indicator	Substantive factor loading		Method factor loading	
		($R1$)	$R1^2$	($R2$)	$R2^2$
Website quality	SEC	0.694***	0.482	0.035	0.000
	NAV	0.774***	0.599	0.010	0.000
	VIS	0.880***	0.774	-0.039	0.000
Transaction convenience	TC1	0.887***	0.787	-0.000	0.000
	TC2	0.918***	0.843	-0.038	0.000
	TC3	0.875***	0.766	-0.020	0.000
	TC4	0.832***	0.692	0.058	0.000
Initiator reputation	REP1	0.686***	0.471	0.226	0.000
	REP2	0.654***	0.428	0.065	0.000
	REP 3	0.974***	0.949	-0.289	0.010
	REP 4	0.923***	0.852	-0.034	0.000
	REP 5	0.845***	0.714	0.070	0.003
Project popularity	POP1	0.871***	0.759	0.003	0.000
	POP2	0.958***	0.918	-0.039	0.001
	POP3	0.869***	0.755	0.038	0.001
Project content quality	CON1	0.935***	0.874	-0.004	0.000
	CON2	0.939***	0.882	0.003	0.001
	CON3	0.926***	0.857	0.001	0.000
Empathy	EMP1	0.693***	0.480	-0.080	0.023
	EMP2	0.643***	0.413	0.102	0.003
	EMP3	0.855***	0.731	-0.069	0.005
	EMP4	0.922***	0.850	-0.073	0.000
	EMP5	0.934***	0.872	-0.103	0.004
	EMP6	0.667***	0.445	0.208	0.008
Perceived credibility	CRE1	0.899***	0.808	0.015	0.000
	CRE2	0.874***	0.764	0.042	0.000
	CRE3	0.896***	0.803	-0.023	0.000
	CRE4	0.984***	0.968	-0.082	0.007
	CRE5	0.761***	0.579	0.135	0.014
	CRE6	0.903***	0.815	-0.094	0.006
Intention to donate	INT1	0.829***	0.687	0.125	0.009
	INT2	0.927***	0.859	0.018	0.002
	INT3	0.926***	0.857	-0.152	0.032
Average		0.853	0.737	0.069	0.004

Table VI.
Common method
bias analysis

Notes: *** $p < 0.001$

0.378 to 0.681. According to Hair *et al.* (1998), the VIF values were far below the suggested value of 10. The VIF and tolerance values of this study were also within the more stringent criteria ($VIF < 5$ and tolerance > 0.2) as indicated by Groebner *et al.* (2011). Hence, multicollinearity was not a serious problem of this study.

5.2 Structural model

As illustrated in Figure 2, ten out of the twelve hypotheses were supported ($p < 0.05$), while the remaining two were not significant at the 0.05 level of significance. First, empathy ($\beta = 0.29, p < 0.001$) and perceived credibility ($\beta = 0.33, p < 0.001$) exhibited strong positive effects on intention to donate. They jointly explained 55.8 percent of the variance in intention to donate. Consequently, *H1* and *H2* were supported empirically. Second, in terms of technological characteristics, website quality significantly and positively affected empathy ($\beta = 0.29, p < 0.001$) and perceived credibility ($\beta = 0.17, p < 0.05$), supporting *H3a* and *H3b*. In addition, transaction convenience was associated positively with empathy ($\beta = 0.22, p < 0.001$) and perceived credibility ($\beta = 0.14, p < 0.05$). Therefore, we concluded that *H4a* and *H4b* were supported. Third, findings on impacts of project characteristics were more complicated. Initiator reputation was positively related to perceived credibility ($\beta = 0.16, p < 0.05$), confirming *H5b*. However, the results showed an insignificant path between initiator reputation and empathy. The path from project popularity to empathy ($\beta = 0.12, p < 0.05$) was positive and significant, while project popularity showed no significant influence on perceived credibility. Consequently, *H6a* was supported while *H6b* was not. Project content quality exerted significant and positive impacts on both empathy ($\beta = 0.21, p < 0.01$) and perceived credibility ($\beta = 0.42, p < 0.001$). Therefore, *H7a* and *H7b* were supported. The proportions of variances explained were 49.8 percent for empathy, and 53.7 percent for perceived credibility. Finally, the results indicated that two control variables – altruism ($\beta = 0.15, p < 0.05$) and past donation experience ($\beta = 0.23, p < 0.001$) – significantly affected intention to donate.

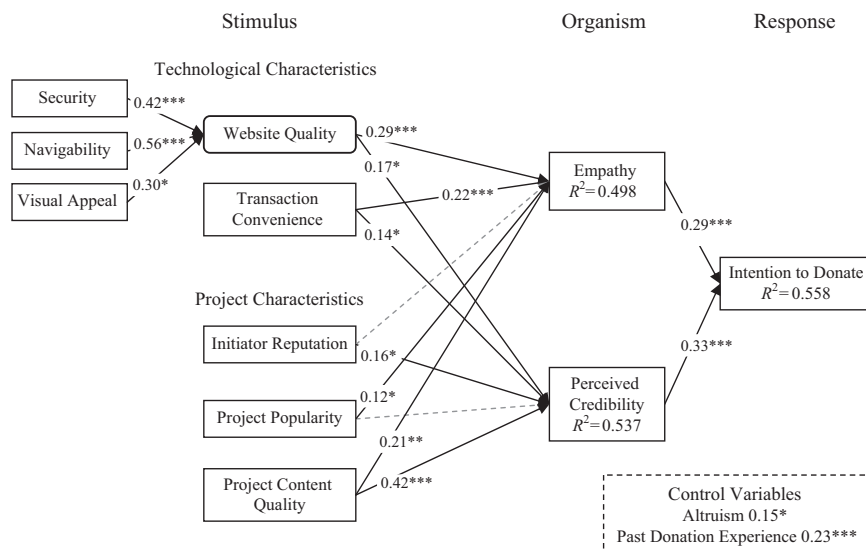


Figure 2.
Structural model

Table VII.
Summary of
structural model

The proposed hypotheses and data analysis results are summarized in Table VII. In conclusion, majority of the hypotheses are supported, except for the impact of initiator reputation on empathy and the influence of project popularity on perceived credibility of a charitable crowdfunding project. Besides, among the 205 respondents, we reported in this study, 157 respondents had donation experience while 48 respondents had no donation experience. In order to check the robustness of our findings, we used SmartPLS 2.0 to test the structural model with 157 samples who had donation experience and compared with the results of analyzing 205 samples. Findings indicated that the results were consistent across the two groups (see Table AIII, we summarized the results of hypotheses test between two groups; in Figure 3, structural model with 157 respondents who had no donation experience was shown).

Hypotheses	Description	Result
H1	Empathy is positively related to intention to donate	Supported
H2	Perceived credibility is positively related to intention to donate	Supported
H3a	Website quality is positively related to empathy	Supported
H3b	Website quality is positively related to perceived credibility	Supported
H4a	Transaction convenience is positively related to perceived credibility	Supported
H4b	Transaction convenience is positively related to empathy	Supported
H5a	Initiator reputation is positively related to empathy	Not supported
H5b	Initiator reputation is positively related to perceived credibility	Supported
H6a	Project popularity is positively related to empathy	Supported
H6b	Project popularity is positively related to perceived credibility	Not supported
H7a	Project content quality is positively related to empathy	Supported
H7b	Project content quality is positively related to perceived credibility	Supported

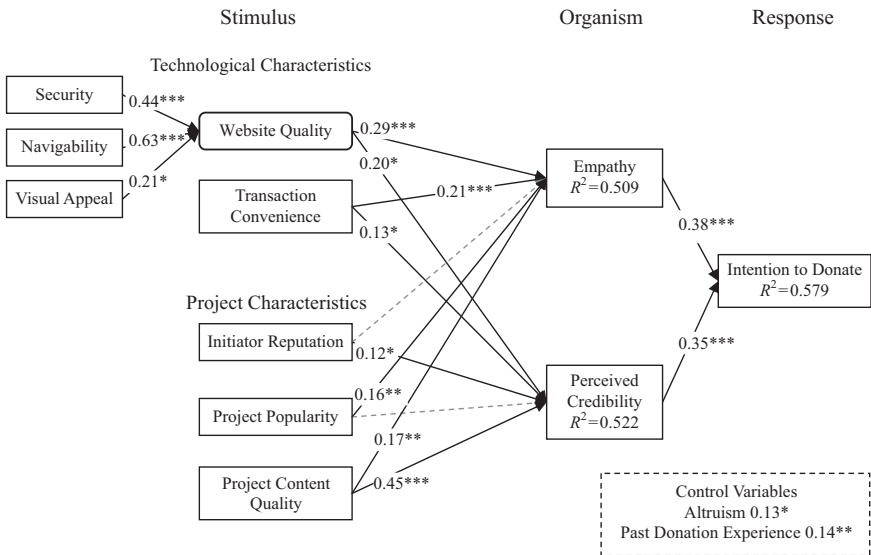


Figure 3.
Structural model with
157 respondents who
had donation
experience

Notes: □ First-order construct; □ second-order construct; → significant relationship; ---→ insignificant relationship. * $p<0.05$; ** $p<0.01$; *** $p<0.001$

6. Discussion and implications

The aim of this study was to develop and test a comprehensive framework that explains individual donation behavior in crowdfunding. Since charitable crowdfunding is gaining popularity as an alternative vehicle to traditional charity, it becomes crucial to understand the drivers of fundraising success or failure. However, relatively little is known about the mechanisms that drive individuals to donate in charitable crowdfunding. This research responds to a call to advance our knowledge of individuals' donation behavior for charitable crowdfunding.

The results indicate that individuals' empathy and the perceived credibility of a charitable crowdfunding project play powerful and additive roles in determining their intention to donate money. Although previous studies have investigated the effects of empathy and perceived credibility on individuals' intention to donate, they mostly have not examined their additive effects; they investigated the effect of either empathy or perceived credibility on intention to donate for charitable crowdfunding (Kim *et al.*, 2016). The results also show that there is no significant correlation between empathy and perceived credibility, justifying the need to distinguish emotional state from cognitive state and to investigate them together in a comprehensive model.

Second, evidence is provided for the positive effects of website quality, transaction convenience, and project content quality on both empathy and perceived credibility. The results associated with the impact of technological characteristics (website quality and transaction convenience) on empathy warrant further discussion. Previous research on empathy has largely regarded it as a personal characteristic (Decety and Jackson, 2006; Hoffman, 2001; Lee *et al.*, 2014) and overlooked how technology (e.g. website quality) stimulates empathy. Our findings add to the literature on empathy by empirically verifying how technology exemplifies empathy. More specifically, website quality in terms of security, navigability, visual appeal, and transaction convenience are key factors that influence individuals' empathy for charitable crowdfunding projects.

Third, it is noteworthy that initiator reputation is positively related to a project's perceived credibility, while project popularity is positively associated with empathy. As comments, likes, and donations made by other individuals are visible in charitable crowdfunding, the initial, pure project popularity will be extended by additional information when a crowdfunding project is ongoing. It is proven that project visitors will be influenced by the additional information representing the behavior of previous visitors to the project, described elsewhere as a herding effect (Beier and Wagner, 2015). Our observation regarding the positive impact of project popularity on empathy further confirms this argument. Contrary to our expectations, initiator reputation did not have a significant impact on empathy, and project popularity did not have a significant impact on its perceived credibility. In addition, we failed to identify a significant effect of social ties with the project initiators on intention to donate, which differs from the findings of other crowdfunding types. For instance, existing studies report that interpersonal relations significantly influence funding behavior in revenue-share crowdfunding, especially at the early stage (Agrawal *et al.*, 2015; Liang and Yuan, 2016). One possible explanation may be that most individuals who donate money in charitable crowdfunding have built weak or no social connections with fund seekers, and they do not expect material reward. Hence, it is not necessary for them to be familiar with the funding seekers when making donation decisions.

6.1 Theoretical implications

In this study, we drew on previous charity and crowdfunding research to investigate the determinants of individual donation behavior in charitable crowdfunding. The results of this research make a few key contributions to the existing body of knowledge on charitable crowdfunding through IS wisdom. This study is among the first to combine and empirically

examine the effects of an emotional state (empathy) and a cognitive state (perceived credibility) on intention to donate in the context of charitable crowdfunding. It contributes to the IS literature by demonstrating that both emotional and cognitive states positively affect individuals' intention to donate. In addition, the findings suggest that there is no significant correlation between empathy and perceived credibility, further proving the importance of distinguishing the emotional state from the cognitive state and investigating their respective influences.

Second, the study identifies technological characteristics (e.g. website quality, transaction convenience) and crowdfunding project characteristics (e.g. initiator reputation, project popularity, information quality) as two environmental cues, and quantifies their influences on emotional and cognitive states (e.g. empathy and perceived credibility). It expands the literature on human-computer interaction by conceptualizing and investigating how technology elicits empathy. This research also contributes to an overall conceptual understanding of the nature and importance of facets of environmental cues in affecting individuals' donation behavior in charitable crowdfunding, mediated by emotional and cognitive states.

Third, while previous research focuses primarily on exploring why people (including both initiators and funders) participate in charitable crowdfunding, the present research contributes to our knowledge by conducting a granular analysis from the perspective of funders. Fourth, the study contributes to the S-O-R model by incorporating both emotional and cognitive states into the "organism" stage, and adapting and verifying it in the charitable crowdfunding context.

6.2 Practical implications

This study also provides important implications for the better organization of charitable crowdfunding projects. We call practitioners' attention to technological design features and project features. More specifically, our research indicates that website quality, transaction convenience, initiator reputation, project popularity, and project content quality should be strategically managed to elicit empathy and perceived credibility, which may induce individuals' intentions to donate money in charitable crowdfunding.

First, the crowdfunding platform and payment system should be carefully designed to reduce the efforts involved in completing donations. For example, a simplified transaction system that enables individuals to manage their donation to crowdfunding projects with minimal effort (i.e. via a small number of clicks) may lead users to believe that accomplishing the donation is more convenience. Enhanced transaction convenience boosts empathy and perceived credibility, thus helping to generate greater intention to donate money.

Second, our results may serve to support managers in their decisions on whether and how to set up a charitable crowdfunding project's content. The findings indicate that project content is the strongest predictor of both empathy and perceived credibility. Hence, providing project content of a high quality (e.g. accurate, up-to-date, well organized) may maximize donations to charitable crowdfunding. In addition, making use of social media (e.g. Weibo, Facebook) and communication tools will help to improve the popularity of a project, which results in enhanced intention to donate.

Third, with regard to individuals who participate in charitable crowdfunding, our work reinforces the prior finding that the majority of them are millennials (Brown, 2011). The millennial generation, who are described as "digital natives" (Fife-Schaw, 2013) who use social media very frequently (Scott, 2016), are reportedly the biggest audience of crowdfunding (Brown, 2011). Therefore, cultivating the online millennial funder is inevitably central to making crowdfunding successful (Riley-Huff *et al.*, 2016). To leverage the power of millennials more effectively, crowdfunding platform operators are encouraged to collect the users' demographic information. By this means, initiators could directly target and harness the millennials while exhibiting, promoting and distributing a crowdfunding project, in their efforts to enhance the possibility of crowdfunding success.

6.3 Limitations

Our study is subject to several limitations. First, our data originated from only two Chinese crowdfunding platforms, namely Weibo and WeChat, which are essentially social media platforms with extensions that enable crowdfunding. It appears that this study yields insights into the Chinese charitable crowdfunding landscape in particular; thus, the generalizability to other crowdfunding platforms and cultures remains in question. Prior research has demonstrated that cultural differences play an important role in individuals' decision making in crowdfunding (Loiacono *et al.*, 2007). Hence, future work can build on the present study by collecting data from other platforms and from regions that have a different cultural background. Another concern with our methodology is that in our respondents, those who have never donated money to charitable crowdfunding projects are included. According to expectation confirmation theory (Bhattacharjee, 2001), intention might change once an individual has funded a project and begun to receive feedback on his or her funding behavior. We thus suggest that future studies compare the model between those who have donation experience and those who have never donated money before, and investigate whether there are differences among the hypotheses.

Second, we derived environmental cues (including technological characteristics and project characteristics) based on prior charity and crowdfunding studies and treated them exclusively as the predictors of empathy and perceived credibility. In fact, complex social mechanisms also exist in charitable crowdfunding. In this study, we examined individuals' social ties with initiators as a control variable. Although various tools are provided to support interaction among these individuals, social ties among them are neglected in the current assessment. In the case of conventional charity, an initiator's reputation is often built through interpersonal interactions. Therefore, social indicators, such as a potential funder's social ties with other participants in charitable crowdfunding should be investigated as part of future research. In addition, initiators may use different methods to interact with potential funders (e.g. using social media or fact-to-face communication). However, in the current study, we do not have information about how initiators communicated with potential funders. Future work could explore how potential funders' trustworthiness toward different communication media affects our model (e.g. adding trustworthiness toward communication media as a moderator). Furthermore, the effects of potential funders' prior knowledge (Marom and Sade, 2013) and of government regulation (Lee *et al.*, 2016) on crowdfunding projects have been identified as important factors in crowdfunding, but are overlooked in our study. The effects of the above-mentioned variables remain open for future research.

Third, charitable crowdfunding is not limited to donations to individuals in disadvantaged areas, nonprofit organizations (e.g. the documentary film industry) (Hobbs *et al.*, 2016; Sørensen, 2012) are important initiators as well. Previous research illustrates that donation behaviors toward organizations may be influenced by different factors, such as administrative wastefulness (Wash, 2013). This suggests that individuals' donation behavior in charitable crowdfunding regarding individuals and organizations may be subject to different threats. Moreover, although funders in charitable crowdfunding do not expect material rewards in exchange for their donations (Giudici *et al.*, 2012), they may anticipate a social reward instead (e.g. acknowledgments) (Leimeister and Zogaj, 2013), which deserves further investigation in the future.

References

- Agrawal, A., Catalini, C. and Goldfarb, A. (2015), "Crowdfunding: geography, social networks, and the timing of investment decisions", *Journal of Economics & Management Strategy*, Vol. 24 No. 2, pp. 253-274.
- Allison, T.H., Davis, B.C., Webb, J.W. and Short, J.C. (2017), "Persuasion in crowdfunding: an elaboration likelihood model of crowdfunding performance", *Journal of Business Venturing*, Vol. 32 No. 6, pp. 707-725.

- Althoff, T. and Leskovec, J. (2015), "Donor retention in online crowdfunding communities: a case study of donorschoose.org", *Proceedings of the 24th International Conference on World Wide Web, International World Wide Web Conferences Steering Committee*, pp. 34-44.
- Batson, C.D., Fultz, J. and Schoenrade, P.A. (1987), "Distress and empathy: two qualitatively distinct vicarious emotions with different motivational consequences", *Journal of Personality*, Vol. 55 No. 1, pp. 19-39.
- Batson, C.D., Klein, T.R., Highberger, L. and Shaw, L.L. (1995), "Immorality from empathy-induced altruism: when compassion and justice conflict", *Journal of Personality and Social Psychology*, Vol. 68 No. 6, pp. 1042-1054.
- Beaulieu, T., Sarker, S. and Sarker, S. (2015), "A conceptual framework for understanding crowdfunding", *Communications of the Association for Information Systems*, Vol. 37 No. 1, pp. 1-31.
- Beier, M. and Wagner, K. (2015), "Crowdfunding success: a perspective from social media and e-commerce", *Thirty-Sixth International Conference on Information Systems, Fort Worth, TX, December 13-16*.
- Belleflamme, P., Lambert, T. and Schwienbacher, A. (2013), "Individual crowdfunding practices", *Venture Capital*, Vol. 15 No. 4, pp. 313-333.
- Belleflamme, P., Lambert, T. and Schwienbacher, A. (2014), "Crowdfunding: tapping the right crowd", *Journal of Business Venturing*, Vol. 29 No. 5, pp. 585-609.
- Bennett, R. (2009), "Impulsive donation decisions during online browsing of charity websites", *Journal of Consumer Behaviour*, Vol. 8 No. 2, pp. 116-134.
- Bhattacharjee, A. (2001), "Understanding information systems continuance: an expectation-confirmation model", *MIS Quarterly*, Vol. 25 No. 3, pp. 351-370.
- Bickart, B. and Schindler, R.M. (2001), "Internet forums as influential sources of consumer information", *Journal of Interactive Marketing*, Vol. 15 No. 3, pp. 31-40.
- Brown, S. (2011), "Measures of shape: skewness and kurtosis", available at: <http://BrownMath.com/stat/shape.htm> (accessed September 18, 2017).
- Burtch, G., Ghose, A. and Wattal, S. (2013), "Cultural differences and geography as determinants of online pro-social lending", *MIS Quarterly*, Vol. 38 No. 3, pp. 773-794.
- Burtch, G., Ghose, A. and Wattal, S. (2014), "An experiment in crowdfunding: assessing the role and impact of transaction-level information controls", *Proceedings of the 2014 International Conference on Information Systems, Auckland, December 14-17*.
- Burtch, G., Ghose, A. and Wattal, S. (2015), "The hidden cost of accommodating crowdfunder privacy preferences: a randomized field experiment", *Management Science*, Vol. 61 No. 5, pp. 949-962.
- Capota, K., Van Hout, M. and Van Der Geest, T. (2007), "Measuring the emotional impact of websites: a study on combining a dimensional and discrete emotion approach in measuring visual appeal of university websites", in Koskinen, I., Kurvinen, E., Mattelmäki, T. and Vaajakallio, K. (Eds), *Proceedings of the 2007 Conference on Designing Pleasurable Products and Interfaces, ACM, Helsinki*, pp. 135-147.
- Castillo, C., Mendoza, M. and Poblete, B. (2011), "Information credibility on twitter", in Ghinita, G. and Punera, K. (Eds), *Proceedings of the 20th International Conference on World Wide Web, ACM, Hyderabad*, pp. 675-684.
- Cecere, G., Le Guel, F. and Rochelandet, F. (2017), "Crowdfunding and social influence: an empirical investigation", *Applied Economics*, Vol. 49 No. 57, pp. 1-12.
- Centefelli, R.T. and Bassellier, G. (2009), "Interpretation of formative measurement in information systems research", *MIS Quarterly*, Vol. 33 No. 4, pp. 689-707.
- Chang, Y.-T., Yu, H. and Lu, H.-P. (2015), "Persuasive messages, popularity cohesion, and message diffusion in social media marketing", *Journal of Business Research*, Vol. 68 No. 4, pp. 777-782.
- Chen, X.-P., Hui, C. and Sego, D.J. (1998), "The role of organizational citizenship behavior in turnover: conceptualization and preliminary tests of key hypotheses", *Journal of Applied Psychology*, Vol. 83 No. 6, pp. 922-931.

- Cheung, M.Y., Luo, C., Sia, C.L. and Chen, H. (2009), "Credibility of electronic word-of-mouth: informational and normative determinants of on-line consumer recommendations", *International Journal of Electronic Commerce*, Vol. 13 No. 4, pp. 9-38.
- Chin, W.W. (1998), "The partial least squares approach to structural equation modeling", *Modern Methods for Business Research*, Vol. 295 No. 2, pp. 295-336.
- Chin, W.W., Marcolin, B.L. and Newsted, P.R. (2003), "A partial least squares latent variable modeling approach for measuring interaction effects: results from a Monte Carlo simulation study and an electronic-mail emotion/adoption study", *Information Systems Research*, Vol. 14 No. 2, pp. 189-217.
- Chiu, C.-M., Hsu, M.-H. and Wang, E.T. (2006), "Understanding knowledge sharing in virtual communities: an integration of social capital and social cognitive theories", *Decision Support Systems*, Vol. 42 No. 3, pp. 1872-1888.
- Choy, K. and Schlagwein, D. (2016), "Crowdsourcing for a better world: on the relation between IT affordances and donor motivations in charitable crowdfunding", *Information Technology & People*, Vol. 29 No. 1, pp. 221-247.
- Crawford, M.B. (2015), "Introduction, attention as a cultural problem", in Farrar, S.A.G. (Ed.), *The World Beyond Your Head: On Becoming an Individual in an Age of Distraction*, Macmillan, New York, NY, pp. 3-27.
- Crowdsourcing, L. (2012), "Crowdfunding industry report: market trends, composition and crowdfunding platforms", Crowdfund Insider, available at: www.crowdfundinsider.com/2014/03/34255-crowdfunding-fraud-big-threat/ (accessed September 18, 2017).
- De Vries, L., Gensler, S. and Leeﬂang, P.S. (2012), "Popularity of brand posts on brand fan pages: an investigation of the effects of social media marketing", *Journal of Interactive Marketing*, Vol. 26 No. 2, pp. 83-91.
- Decety, J. and Jackson, P.L. (2006), "A social-neuroscience perspective on empathy", *Current Directions in Psychological Science*, Vol. 15 No. 2, pp. 54-58.
- Dodds, W.B., Monroe, K.B. and Grewal, D. (1991), "Effects of price, brand, and store information on buyers' product evaluations", *Journal of Marketing Research*, Vol. 28 No. 3, August, pp. 307-319.
- Du, Q., Fan, W., Qiao, Z., Wang, G., Zhang, X. and Zhou, M. (2015), "Money talks: a predictive model on crowdfunding success using project description", *Proceedings of the Twenty-first Americas Conference on Information Systems, Puerto Rico, August 13-15*.
- Eisenberg, N. and Miller, P.A. (1987), "The relation of empathy to prosocial and related behaviors", *Psychological Bulletin*, Vol. 101 No. 1, pp. 91-119.
- Everard, A. and Galletta, D.F. (2005), "How presentation flaws affect perceived site quality, trust, and intention to purchase from an online store", *Journal of Management Information Systems*, Vol. 22 No. 3, pp. 56-95.
- Feller, J., Gleasure, R. and Treacy, S. (2013), "Crowdfunding: past research, future directions", *European Conference on Information Systems, Utrecht, June 6-8*.
- Fife-Schaw, C. (2013), "Statistics FAQ", available at: www.surrey.ac.uk/psychology/current/statistics/index.htm (accessed September 18, 2017).
- Fornell, C. and Larcker, D.F. (1981), "Evaluating structural equation models with unobservable variables and measurement error", *Journal of Marketing Research*, Vol. 18 No. 1, pp. 39-50.
- Freeman, K.S. and Spyridakis, J.H. (2004), "An examination of factors that affect the credibility of online health information", *Technical Communication*, Vol. 51 No. 2, pp. 239-263.
- Fung, R. and Lee, M. (1999), "EC-trust (trust in electronic commerce): exploring the antecedent factors", in And, W.D.H. and Nazareth, D.L. (Eds), *Proceedings of Fifth Americas Conference on Information Systems, ACM, New York, NY*, pp. 517-519.
- Galak, J., Small, D. and Stephen, A.T. (2011), "Microfinance decision making: a field study of prosocial lending", *Journal of Marketing Research*, Vol. 48, pp. S130-S137.
- Gerber, E.M. and Hui, J. (2013), "Crowdfunding: motivations and deterrents for participation", *ACM Transactions on Computer-Human Interaction*, Vol. 20 No. 6, pp. 1-33.

- Gerber, E.M., Hui, J.S. and Kuo, P.-Y. (2012), "Crowdfunding: why people are motivated to post and fund projects on crowdfunding platforms", in Poltrock, S. and Simone, C. (Eds), *Proceedings of the International Workshop on Design, Influence, and Social Technologies: Techniques, Impacts and Ethics*, ACM, New York, NY, pp. 1-11.
- Gerdes, K.E. (2011), "Empathy, sympathy, and pity: 21st-century definitions and implications for practice and research", *Journal of Social Service Research*, Vol. 37 No. 3, pp. 230-241.
- Gierczak, M., Bretschneider, U. and Leimeister, J.M. (2014), "Is all that glitters gold? Exploring the effects of perceived risk on backing behavior in reward-based crowdfunding", *Proceedings of the Thirty-fifth International Conference on Information Systems, Auckland, December 14-17*.
- Giudici, G., Nava, R., Rossi Lamastra, C. and Verecondo, C. (2012), "Crowdfunding: the new frontier for financing entrepreneurship?", available at: <https://ssrn.com/abstract=2157429> (accessed September 18, 2017).
- Gleasure, R. and Feller, J. (2016a), "Does heart or head rule donor behaviors in charitable crowdfunding markets?", *International Journal of Electronic Commerce*, Vol. 20 No. 4, pp. 499-524.
- Gleasure, R. and Feller, J. (2016b), "Emerging technologies and the democratisation of financial services: a metatriangulation of crowdfunding research", *Information and Organization*, Vol. 26 No. 4, pp. 101-115.
- Greenberg, M.D. and Gerber, E.M. (2014), "Learning to fail: experiencing public failure online through crowdfunding", in Schmidt, A. and Grossman, T. (Eds), *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, ACM, Toronto, pp. 581-590.
- Gregg, D.G. and Walczak, S. (2010), "The relationship between website quality, trust and price premiums at online auctions", *Electronic Commerce Research*, Vol. 10 No. 1, pp. 1-25.
- Groebner, D.F., Shannon, P.W., Fry, P.C. and Smith, K.D. (2011), *Business Statistics: A Decision Making Approach*, Prentice Hall, UpperSaddle River, NJ.
- Guo, Z., Huang, J., He, J., Hei, X. and Wu, D. (2013), "Unveiling the patterns of video tweeting: a sina weibo-based measurement study", in Roughan, M. and Chang, R. (Eds) *Passive and Active Measurement*, Springer, Berlin, pp. 166-175.
- Gvili, Y. and Levy, S. (2016), "Antecedents of attitudes toward eWOM communication: differences across channels", *Internet Research*, Vol. 26 No. 5, pp. 1030-1051.
- Haas, P., Blohm, I. and Leimeister, J.M. (2014), "An empirical taxonomy of crowdfunding intermediaries", *Proceedings of the Thirty-fifth International Conference on Information Systems, Auckland, December 14-17*.
- Hair, J.F., Anderson, R.E., Tatham, R.L. and Black, W.C. (1998), *Multivariate Data Analysis*, Upper Saddle River, NJ.
- Hair, J.F., Black, W.C., Babin, B.J., Anderson, R.E. and Tatham, R.L. (2006), *Multivariate Data Analysis*, Prentice Hall, Pearson Prentice Hall, Upper Saddle River, NJ.
- Hobbs, J., Grigore, G. and Molesworth, M. (2016), "Success in the management of crowdfunding projects in the creative industries", *Internet Research*, Vol. 26 No. 1, pp. 146-166.
- Hoffman, D.L., Novak, T.P. and Peralta, M. (1999), "Building consumer trust online", *Communications of the ACM*, Vol. 42 No. 4, pp. 80-85.
- Hoffman, M.L. (2001), *Empathy and Moral Development: Implications for Caring and Justice*, Cambridge University Press, New York, NY.
- Huber, G.P. (1990), "A theory of the effects of advanced information technologies on organizational design, intelligence, and decision making", *Academy of Management Review*, Vol. 15 No. 1, pp. 47-71.
- Jarvenpaa, S.L., Tractinsky, N. and Saarinen, L. (1999), "Consumer trust in an internet store: a cross-cultural validation", *Journal of Computer-Mediated Communication*, Vol. 5 No. 2, pp. 1-35.
- Jiang, Z., Chan, J., Tan, B.C. and Chua, W.S. (2010), "Effects of interactivity on website involvement and purchase intention", *Journal of the Association for Information Systems*, Vol. 11 No. 1, pp. 34-59.

- Jih, W.-J.K. (2007), "Effects of consumer-perceived convenience on shopping intention in mobile commerce: an empirical study", *International Journal of E-Business Research*, Vol. 3 No. 4, pp. 33-48.
- Jin, B., Yong Park, J. and Kim, J. (2008), "Cross-cultural examination of the relationships among firm reputation, e-satisfaction, e-trust, and e-loyalty", *International Marketing Review*, Vol. 25 No. 3, pp. 324-337.
- Jin, S.-A. A. and Phua, J. (2014), "Following celebrities' tweets about brands: the impact of twitter-based electronic word-of-mouth on consumers' source credibility perception, buying intention, and social identification with celebrities", *Journal of Advertising*, Vol. 43 No. 2, pp. 181-195.
- Jøsang, A., Ismail, R. and Boyd, C. (2007), "A survey of trust and reputation systems for online service provision", *Decision Support Systems*, Vol. 43 No. 2, pp. 618-644.
- Kang, M., Kang, M., Gao, Y., Gao, Y., Wang, T., Wang, T., Zheng, H. and Zheng, H. (2016), "Understanding the determinants of funders' investment intentions on crowdfunding platforms: a trust-based perspective", *Industrial Management & Data Systems*, Vol. 116 No. 8, pp. 1800-1819.
- Kickstarter (2013), "Q&A", available at: www.kickstarter.com/blog/2011-the-stats (accessed September 18, 2017).
- Kim, C., Mirusmonov, M. and Lee, I. (2010), "An empirical examination of factors influencing the intention to use mobile payment", *Computers in Human Behavior*, Vol. 26 No. 3, pp. 310-322.
- Kim, J.G., Kong, H.K., Karahalios, K., Fu, W.-T. and Hong, H. (2016), "The power of collective endorsements: credibility factors in medical crowdfunding campaigns", in Kaye, J., Druin, A., Lampe, C., Morris, D. and Pablo, J. (Eds), *Proceedings of Computer Human Interaction, ACM, San Jose, CA*, pp. 4538-4549.
- Kuo, Y.-F. and Wu, C.-H. (2014), "Understanding the drivers of sponsors' intentions in online crowdfunding: a model development", in Tao, Y.-H., Yang, H.-C., Ting, I.-H., Steinbauer, M., Khalil, I. and Anderst-Kotsis, G. (Eds), *Proceedings of the 12th International Conference on Advances in Mobile Computing and Multimedia, ACM, New York, NY*, pp. 433-438.
- Kuppuswamy, V. and Bayus, B.L. (2015), "Crowdfunding creative ideas: the dynamics of project backers in Kickstarter", UNC Kenan-Flagler research paper, University of North Carolina Kenan-Flagler Business School, Chapel Hill, NC, available at: <http://ssrn.com/abstract=223476> (accessed September 18, 2017).
- Lacan, C., Lacan, C., Desmet, P. and Desmet, P. (2017), "Does the crowdfunding platform matter? Risks of negative attitudes in two-sided markets", *Journal of Consumer Marketing*, Vol. 34 No. 6, pp. 472-479.
- Lambert, T. and Schwenbacher, A. (2010), "An empirical analysis of crowdfunding", Social Science Research Network, University de Louvain, Leuven, available at: <http://ssrn.com/abstract=1578175>.
- Lee, C.H., Zhao, J.L. and Hassna, G. (2016), "Government-incentivized crowdfunding for one-belt, one-road enterprises: design and research issues", *Financial Innovation*, Vol. 2 No. 2, pp. 1-14.
- Lee, J., Park, D.-H. and Han, I. (2008), "The effect of negative online consumer reviews on product attitude: an information processing view", *Electronic Commerce Research and Applications*, Vol. 7 No. 3, pp. 341-352.
- Lee, S., Winterich, K.P. and Ross, W.T. (2014), "I'm moral, but I won't help you: the distinct roles of empathy and justice in donations", *Journal of Consumer Research*, Vol. 41 No. 3, pp. 678-696.
- Leimeister, J.M. and Zogaj, S. (2013), "Neue Arbeitsorganisation durch Crowdsourcing: Eine Literaturstudie", Arbeitspapier der Hans-Böckler-Stiftung, Reihe Arbeit und Soziales Nr. 287, Hans Böckler Foundation, Düsseldorf, pp. 1-112.
- Li, Y.-Z., He, T.-L., Song, Y.-R., Yang, Z. and Zhou, R.-T. (2017), "Factors impacting donors' intention to donate to charitable crowd-funding projects in China: a UTAUT-based model", *Information, Communication & Society*, Vol. 20 No. 3, pp. 1-12.
- Liang, H., Saraf, N., Hu, Q. and Xue, Y. (2007), "Assimilation of enterprise systems: the effect of institutional pressures and the mediating role of top management", *MIS Quarterly*, Vol. 31 No. 1, pp. 59-87.

- Liang, Y.E. and Yuan, S.-T.D. (2016), "Predicting investor funding behavior using crunchbase social network features", *Internet Research*, Vol. 26 No. 1, pp. 74-100.
- Liu, L., Suh, A. and Wagner, C. (2017), "Donation behavior in online micro charities: an investigation of charitable crowdfunding projects", *Proceedings of the 50th Hawaii International Conference on System Sciences*, Big Island, HI, pp. 843-852.
- Loiacono, E.T., Watson, R.T. and Goodhue, D.L. (2007), "WebQual: an instrument for consumer evaluation of web sites", *International Journal of Electronic Commerce*, Vol. 11 No. 3, pp. 51-87.
- Lu, C.-T., Xie, S., Kong, X. and Yu, P.S. (2014), "Inferring the impacts of social media on crowdfunding", *Proceedings of the 7th ACM International Conference on Web Search and Data Mining*, New York, NY, pp. 573-582.
- McCroskey, J.C. and Teven, J.J. (1999), "Goodwill: a reexamination of the construct and its measurement", *Communications Monographs*, Vol. 66 No. 1, pp. 90-103.
- McGowan, M. (2012), "Crowdfunding and nonprofits: creating successful campaigns", available at: <http://megmdesigns.com/img/NonProfit%20Fundraising.pdf> (accessed September 18, 2017).
- Marom, D. and Sade, O. (2013), "Are the life and death of an early stage venture indeed in the power of the tongue? Lessons from online crowdfunding pitches", available at: <http://ssrn.com/abstract=2255707> (accessed September 18, 2017).
- Marsden, P. and Lin, N. (1982), *Social Structure and Networks Analysis*, Sage, Beverley Hills, CA.
- Massolution, C. (2015), "Crowdfunding industry report: market trends, composition and crowdfunding platforms", Crowdfund Insider, available at: www.crowdfundinsider.com/2014/03/34255-crowdfunding-fraud-big-threat/ (accessed September 18, 2017).
- Meng, R., Kang, M., Wang, T. and Zheng, H. (2016), "Understanding the role of commitments in explaining crowdfunding investing willingness: antecedents and consequences", *Proceedings of the 20th Pacific Asia Conference on Information Systems*, Chiayi, June 27-July 1.
- Merriam-Webster (2015), "Charity", available at: www.merriam-webster.com/dictionary/charity (accessed September 18, 2017).
- Moisseyev, A. (2013), *Effect of Social Media on Crowdfunding Project Results*, University of Nebraska, Lincoln.
- Mollick, E. (2012), "The dynamics of crowdfunding: determinants of success and failure", *Social Science Research Network*, Rochester, NY.
- Mollick, E. (2014), "The dynamics of crowdfunding: an exploratory study", *Journal of Business Venturing*, Vol. 29 No. 1, pp. 1-16.
- Moqri, M. and Bandyopadhyay, S. (2016), "Please share! Online word of mouth and charitable crowdfunding", *Proceedings of the 22nd Americas Conference on Information Systems*, San Diego, CA.
- Mummalaneni, V. (2005), "An empirical investigation of web site characteristics, consumer emotional states and on-line shopping behaviors", *Journal of Business Research*, Vol. 58 No. 4, pp. 526-532.
- Murillo, D.E., Kang, J. and Yoon, S. (2016), "Factors influencing pro-social consumer behavior through non-profit organizations", *Internet Research*, Vol. 26 No. 3, pp. 626-643.
- O'Keefe, D.J. (2015), *Persuasion: Theory and Research*, Sage, Newbury Park, CA.
- Park, C.-H. and Kim, Y.-G. (2003), "Identifying key factors affecting consumer purchase behavior in an online shopping context", *International Journal of Retail & Distribution Management*, Vol. 31 No. 1, pp. 16-29.
- Podsakoff, P.M. and Organ, D.W. (1986), "Self-reports in organizational research: problems and prospects", *Journal of Management*, Vol. 12 No. 4, pp. 531-544.
- Podsakoff, P.M., MacKenzie, S.B., Lee, J.-Y. and Podsakoff, N.P. (2003), "Common method biases in behavioral research: a critical review of the literature and recommended remedies", *Journal of Applied Psychology*, Vol. 88 No. 5, pp. 879-903.

- Rick, S.I., Cryder, C.E. and Loewenstein, G. (2008), "Tightwads and spendthrifts", *Journal of Consumer Research*, Vol. 34 No. 6, pp. 767-782.
- Riley-Huff, D.A., Herrera, K., Ivey, S. and Harry, T. (2016), "Crowdfunding in libraries, archives and museums", *The Bottom Line*, Vol. 29 No. 2, pp. 67-85.
- Ringle, C.M., Wende, S. and Will, A. (2005), "SmartPLS 2.0 (beta)", SmartPLS, Hamburg.
- Salganik, M.J., Dodds, P.S. and Watts, D.J. (2006), "Experimental study of inequality and unpredictability in an artificial cultural market", *Science*, Vol. 311 No. 5762, pp. 854-856.
- Salisbury, W.D., Pearson, R.A., Pearson, A.W. and Miller, D.W. (2001), "Perceived security and world wide web purchase intention", *Industrial Management & Data Systems*, Vol. 101 No. 4, pp. 165-177.
- Scott, A. (2016), "How to reach the millennial generation with social media", available at: www.postplanner.com/how-to-reach-the-millennial-generation-with-social-media/ (accessed September 18, 2017).
- Shier, M.L. and Handy, F. (2012), "Understanding online donor behavior: the role of donor characteristics, perceptions of the internet, website and program, and influence from social networks", *International Journal of Nonprofit and Voluntary Sector Marketing*, Vol. 17 No. 3, pp. 219-230.
- Sørensen, I.E. (2012), "Crowdsourcing and outsourcing: the impact of online funding and distribution on the documentary film industry in the UK", *Media, Culture & Society*, Vol. 34 No. 6, pp. 726-743.
- Steele, W.R., Schreiber, G.B., Guiltinan, A., Nass, C., Glynn, S.A., Wright, D.J., Kessler, D., Schlumpf, K.S., Tu, Y. and Smith, J.W. (2008), "The role of altruistic behavior, empathetic concern, and social responsibility motivation in blood donation behavior", *Transfusion*, Vol. 48 No. 1, pp. 43-54.
- Sura, S., Ahn, J. and Lee, O. (2017), "Factors influencing intention to donate via social network site (SNS): from Asian's perspective", *Telematics and Informatics*, Vol. 34 No. 1, pp. 164-176.
- Tan, X., Lu, Y. and Tan, Y. (2016), "Why should I donate? Examining the effects of reputation, peer influence, and popularity on charitable giving over social media platforms", available at: <https://ssrn.com/abstract=2820219> (accessed September 18, 2017).
- Tanaka, K.G. and Volda, A. (2016), "Legitimacy work: invisible work in philanthropic crowdfunding", *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, ACM, pp. 4550-4561.
- Wan, J., Lu, Y., Wang, B. and Zhao, L. (2017), "How attachment influences users' willingness to donate to content creators in social media: a socio-technical systems perspective", *Information & Management*, Vol. 54 No. 7, pp. 837-850.
- Wash, R. (2013), "The value of completing crowdfunding projects", *Seventh International AAAI Conference on Weblogs and Social Media*, Boston, MA.
- Wells, J.D., Parboteeah, V. and Valacich, J.S. (2011), "Online impulse buying: understanding the interplay between consumer impulsiveness and website quality", *Journal of the Association for Information Systems*, Vol. 12 No. 1, pp. 32-56.
- Westerman, D., Spence, P.R. and Van Der Heide, B. (2012), "A social network as information: the effect of system generated reports of connectedness on credibility on Twitter", *Computers in Human Behavior*, Vol. 28 No. 1, pp. 199-206.
- Whitman, A. (1972), "Basil O'Connor, polio crusader, dies", *The New York Times*, March 10, 1972, p. 40.
- Williams, L.J., Edwards, J.R. and Vandenberg, R.J. (2003), "Recent advances in causal modeling methods for organizational and management research", *Journal of Management*, Vol. 29 No. 6, pp. 903-936.
- Xu, J.D., Benbasat, I. and Cenfetelli, R.T. (2013), "Integrating service quality with system and information quality: an empirical test in the e-service context", *MIS Quarterly*, Vol. 37 No. 3, pp. 777-794.
- Yang, T. (2013), *Annual Report on China's Philanthropy Development*, Social Sciences Academic Press, Beijing.

- Zhao, Q., Chen, C.-D., Wang, J.-L. and Chen, P.-C. (2017), "Determinants of backers' funding intention in crowdfunding: Social exchange theory and regulatory focus", *Telematics and Informatics*, Vol. 34 No. 1, pp. 370-384.
- Zheng, H., Hung, J.-L., Qi, Z. and Xu, B. (2016), "The role of trust management in reward-based crowdfunding", *Online Information Review*, Vol. 40 No. 1, pp. 97-118.
- Zhou, M.J., Lu, B., Fan, W.P. and Wang, G.A. (2016), "Project description and crowdfunding success: an exploratory study", *Information Systems Frontiers*, Vol. 20 No. 2, pp. 259-274, available at <https://doi.org/10.1007/s10796-016-9723-1> (accessed September 18, 2017).

Further reading

- Boeuf, B., Darveau, J. and Legoux, R. (2014), "Financing creativity: crowdfunding as a new approach for theatre projects", *Arts Funding*, Vol. 16 No. 3, pp. 33-48.
- Cheung, C.K. and Chan, C.M. (2000), "Social-cognitive factors of donating money to charity, with special attention to an international relief organization", *Evaluation and Program Planning*, Vol. 23 No. 2, pp. 241-253.
- Choy, K. and Schlagwein, D. (2015), "IT affordances and donor motivations in charitable crowdfunding: the 'Earthship Kapita' case", *Proceedings of the Twenty-Third European Conference on Information Systems, Münster, May 27-29*.
- Hibbert, S., Smith, A., Davies, A. and Ireland, F. (2007), "Guilt appeals: persuasion knowledge and charitable giving", *Psychology & Marketing*, Vol. 24 No. 8, pp. 723-742.
- Michel, G. and Rieunier, S. (2012), "Nonprofit brand image and typicality influences on charitable giving", *Journal of Business Research*, Vol. 65 No. 5, pp. 701-707.
- Pitschner, S. and Pitschner-Finn, S. (2014), "Non-profit differentials in crowd-based financing: evidence from 50,000 campaigns", *Economics Letters*, Vol. 123 No. 3, pp. 391-394.
- Sana, M.S. (2014), "Donating behaviour in the non-profit marketing context: an empirical study based on the identity theory model", *Proceedings of the International Conference "Marketing from Information to Decision"*, Babes Bolyai University, EBSCO (Business Source Complete), pp. 230-247.
- Teah, M., Lwin, M. and Cheah, I. (2014), "Moderating role of religious beliefs on attitudes towards charities and motivation to donate", *Asia Pacific Journal of Marketing and Logistics*, Vol. 26 No. 5, pp. 738-760.
- Venable, B.T., Rose, G.M., Bush, V.D. and Gilbert, F.W. (2005), "The role of brand personality in charitable giving: an assessment and validation", *Journal of the Academy of Marketing Science*, Vol. 33 No. 3, pp. 295-312.
- Wymer, W. and Drollinger, T. (2015), "Charity appeals using celebrity endorsers: celebrity attributes most predictive of audience donation intentions", *VOLUNTAS: International Journal of Voluntary and Nonprofit Organizations*, Vol. 26 No. 6, pp. 2694-2717.

Dimension	Factors	References
Cognitive dimension	Perceived credibility	Choy and Schlagwein (2016), Kim <i>et al.</i> (2016)
	Legitimacy of the fundraiser	Tanaka and Volda (2016)
	Sense of personal impact, trust	Althoff and Leskovec (2015)
	Establish trust	McGowan (2012)
	Perceived transaction risk	Lacan <i>et al.</i> (2017)
	Perceived risk; trust; commitment	Zhao <i>et al.</i> (2017)
	Trust	Meng <i>et al.</i> (2016)
	perceived risk	Kuo and Wu (2014)
	Trust beliefs: calculus trust, relationship trust	Kang <i>et al.</i> (2016)
	Distrust of creators' use of funds	Gerber and Hui (2013)
	Issue-related information	Allison <i>et al.</i> (2017)
	Source credibility (competence), argument quality	Du <i>et al.</i> (2015)
	Perceived risks (controlled variable: trust, reputation)	Gierczak <i>et al.</i> (2014)
	Creditworthiness (i.e. success experience, investment in others)	Zheng <i>et al.</i> (2016)
	Trustworthy cues (project owner traits), content of project description	Zhou <i>et al.</i> (2016)
	General attitude toward online donation	Sura <i>et al.</i> (2017)
	Information control	Burtch <i>et al.</i> (2014)
Emotional dimension	Feelings of sympathy and empathy toward the cause; feelings of guilt for not giving	Gerber <i>et al.</i> (2012)
	Individual-intrinsic motivations (e.g. feelings of empathy, sympathy or nostalgia; personal enjoyment and satisfaction), social-intrinsic motivations (e.g. sense of belonging to a team or community)	Choy and Schlagwein (2016)
	Warm glow (i.e. positive emotions; pleasure)	Cecere <i>et al.</i> (2017), Gleasure and Feller (2016a)
	Feelings of connectedness to a community with similar interests and ideals	Choy and Schlagwein (2016)
	Similarities in occupation and gender, sharing a first initial	Galak <i>et al.</i> (2011)
	Cultural similarities/differences between lender and borrower countries, geographical proximity	Burtch <i>et al.</i> (2013)
	Technological dimension	Website characteristics (platform-cognition affordances; platform-action affordances)
Choy and	Schlagwein (2016)	Burtch <i>et al.</i> (2015)
	Facilitating anonymity	Sura <i>et al.</i> (2017)
	Internet technological features, SNS features	Burtch <i>et al.</i> (2014)
	Information control	Kuo and Wu (2014)
	Website service quality	Kang <i>et al.</i> (2016)
	Perceived accreditation, structural assurance, third-party seal	Project-cognition affordances; project-action affordances
Choy and	Crowdfunding project characteristics	
	Schlagwein (2016)	Galak <i>et al.</i> (2011)
	Individual vs group fundraisers	Sura <i>et al.</i> (2017)
	Charity project, charity organization	Gleasure and Feller (2016a)
	Donations to organizations: fundraising target; rate of donation; level of disclosure	

(continued)

Table AI.
Factors examined in
crowdfunding
literature

Table AI.

Dimension	Factors	References
	Donations to individuals: level of disclosure; campaign imagery; campaign dialogue	Shier and Handy (2012)
	Perception of the organization	Li <i>et al.</i> (2017)
	Facilitating conditions	Wan <i>et al.</i> (2017)
	Emotional attachment	Moqri and Bandyopadhyay (2016)
	Perception of urgency	Tan <i>et al.</i> (2016)
	Popularity effects, high visibility of funders' contributions	Kang <i>et al.</i> (2016)
	Network externality, perceived informativeness	Kang <i>et al.</i> (2016)
	Value congruence, social interaction ties	Kang <i>et al.</i> (2016)

Appendix 2

Constructs	Items	Measures	References
<i>Website quality</i>			
Security	SEC1	While using Weibo/WeChat, I am confident that the information I provide during my donation will not reach inappropriate parties	Gleasure and Feller (2016b)
	SEC2	While using Weibo/WeChat, I believe inappropriate parties cannot deliberately observe the information I provide during my donation	
	SEC3	While using Weibo/WeChat, inappropriate parties will not collect and store the information I provide during my donation	
Navigability	NAV1	Navigating Weibo/WeChat is easy for me	Salisbury <i>et al.</i> (2001)
	NAV2	I find that my interaction with Weibo/WeChat is clear and understandable	
	NAV3	It is easy for me to become skillful at navigating Weibo/WeChat	
Visual appeal	VIS1	Weibo/WeChat is visually pleasing	Loiacono <i>et al.</i> (2007)
	VIS2	Weibo/WeChat displays visually pleasing design	
	VIS3	Weibo/WeChat is visually appealing	
<i>Transaction convenience</i>			
	TC1	It is convenient to pay for the crowdfunding project	Kim <i>et al.</i> (2010)
	TC2	When the crowdfunding project is still open, I am able to pay for it at any time	
	TC3	When the crowdfunding project is still open, I am able to pay for it in any situation	
	TC4	It is not complex to manage my payment to the crowdfunding project	
<i>Initiator reputation</i>			
	REP1	The initiator of this project has an authorized account on Weibo/WeChat	Jarvenpaa <i>et al.</i> (1999)
	REP2	I am familiar with the initiator's Weibo/WeChat account	
	REP3	The initiator of this project is well-known on Weibo/WeChat	
	REP4	The initiator of this project has a good reputation on Weibo/WeChat	
	REP5	The initiator of this project has a reputation of being honest	
<i>Project popularity</i>			
	POP1	Many users of Weibo/WeChat have pressed like on the crowdfunding project	Chang <i>et al.</i> (2015)
	POP2	Many users of Weibo/WeChat have shared the crowdfunding project	
	POP3	Many users of Weibo/WeChat have commented on the crowdfunding project	
<i>Project content quality</i>			
	CON1	Overall, I would give the content quality of the project high marks	Xu <i>et al.</i> (2013)
	CON2	Overall, I would give a high rating in terms of the content quality for the crowdfunding project	
	CON3	In general, high-quality content for the crowdfunding project is provided	
<i>Empathy</i>			
		When you think about the crowdfunding project, to what extent do you feel the following emotions?	Batson <i>et al.</i> (1987)

(continued)

Table AII.
Constructs and
measures

Constructs	Items	Measures	References
	EMP1	Sympathetic	
	EMP2	Warm	
	EMP3	Compassionate	
	EMP4	Soft-hearted	
	EMP5	Tender	
	EMP6	Moved	
<i>Perceived credibility</i>			
	CRE1	The crowdfunding project is believable	McCroskey and Teven (1999)
	CRE2	The crowdfunding project is trustworthy	
	CRE3	The crowdfunding project is competent	
	CRE4	The crowdfunding project is credible	
	CRE5	The crowdfunding project is unbiased	
	CRE6	The crowdfunding project is expert	
<i>Intention to donate</i>			
	INT1	The probability that I would donate money to the crowdfunding project is high	Dodds <i>et al.</i> (1991)
	INT2	My willingness to donate money to the crowdfunding project is high	
	INT3	The likelihood of my donating money to the crowdfunding project is high	
<i>Altruism</i>			
	ALT1	I like helping other people even though it is not required	Chen <i>et al.</i> (1998)
	ALT2	I am always ready to help others	
	ALT3	I am willing to give my time to help others	
<i>Social ties with project initiator</i>			
	ST1	I maintain close social relationships with the project initiator on Weibo/WeChat	Chiu <i>et al.</i> (2006)
	ST2	I spend a lot of time interacting with the project initiator on Weibo/WeChat	
	ST3	I know the project initiator on a personal level	
	ST4	I have frequent communication with the project initiator on Weibo/WeChat	

Table AII.

Hypotheses	Description	Result (<i>n</i> = 205)	Result (<i>n</i> = 157)
<i>H1</i>	Empathy is positively related to intention to donate	Supported (0.29***)	Supported (0.38***)
<i>H2</i>	Perceived credibility is positively related to intention to donate	Supported (0.33***)	Supported (0.35***)
<i>H3a</i>	Website quality is positively related to empathy	Supported (0.29***)	Supported (0.29***)
<i>H3b</i>	Website quality is positively related to perceived credibility	Supported (0.17*)	Supported (0.20*)
<i>H4a</i>	Transaction convenience is positively related to perceived credibility	Supported (0.22***)	Supported (0.21***)
<i>H4b</i>	Transaction convenience is positively related to empathy	Supported (0.14*)	Supported (0.13*)
<i>H5a</i>	Initiator reputation is positively related to empathy	Not supported	Not supported
<i>H5b</i>	Initiator reputation is positively related to perceived credibility	Supported (0.16*)	Supported (0.12*)
<i>H6a</i>	Project popularity is positively related to empathy	Supported (0.12*)	Supported (0.16**)
<i>H6b</i>	Project popularity is positively related to perceived credibility	Not supported	Not supported
<i>H7a</i>	Project content quality is positively related to empathy	Supported (0.21**)	Supported (0.17**)
<i>H7b</i>	Project content quality is positively related to perceived credibility	Supported (0.42***)	Supported (0.45***)

Notes: All respondents, *n* = 205 vs respondents who had donation experience, *n* = 157. **p* < 0.05; ***p* < 0.01; ****p* < 0.001

Table AIII.
Summary of the
hypotheses test
between two groups

Corresponding author

Lili Liu can be contacted at: llili85@nuaa.edu.cn